

# Akshit Rampershad

Turning enterprise operations into scalable AI-native systems

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## EXPERIENCE

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### AI Engineer | Cambrian Lab, San Francisco, CA

May 2026 – Present

- Architected an **end-to-end PCN-MPN matching system using a 4-leg hybrid retriever** (exact, fuzzy, prefix, vector) feeding a bounded LLM ranking pass, with a short-circuit exact-match mechanism that resolves unambiguous hits at 0 LLM cost.
- Calibrated our internal matching system, by integrating **OpenAI text-embedding-3-small** (1536-d) across 4 supply chain fields, using **HNSW top-K search** and a **3 band score+margin policy**, sustaining a **flat 0.36s match latency** on 50K+ corpus.
- Validated hybrid-retrieval candidate matches using Claude Sonnet 5-based confidence scoring to **reduce mismatched-part risk in PCNs, achieving 0 false auto confirmations across 81 validation runs** and automating unambiguous matches.
- Eliminated manual assembly and finished good tracing by building a **recursive "where-used" BOM engine with parent-traversal graph algorithms and cycle-detection guards**, fully automating downstream impact analysis for matched part.
- Built PcnAutoIntake, an **LLM orchestration pipeline** automating 29+ daily supplier change intake via structured extraction, entity resolution and confidence gated human reviews, **cutting PCN analysis from 2hr to 10sec/notice, saving 19hrs/day**.

### Data Engineer / AI Solutions Delivery | 22nd Century Technology Inc. McLean, VA

Sept 2025 – Mar 2026

- Spearheaded the **end-to-end technical delivery of a secure enterprise RAG platform** to accelerate multi-million dollar federal proposal development cycles and intelligence generation workloads across 20K+ SharePoint knowledge corpus.
- Built a **hybrid ingestion engine using Python, PySpark, and Graph API to index a 20K+ SharePoint corpus into a 1,024-dim Qdrant vector database**, using BGE-Large-EN-v1.5 embeddings for section-aware retrieval.
- Designed a **LlamaIndex retrieval pipeline narrowing 30 candidates to 8 through five-field metadata filtering and Cross-Encoder re-ranking**, feeding LangChain prompt orchestration to return RFP-scoped drafts sourced from prior wins.
- Managed project dependencies and technical ambiguity for 4 cross-functional teams through Jira, **driving stakeholder collaboration across AI, engineering, and proposal teams** to deliver structured RAG features across 8 continuous sprints.
- Developed AI adoption runbooks and **led technical enablement sessions with business stakeholders and 14 proposals writers**, improving organizational data literacy and scaling RAG application adoption by 45% across business teams.

### Data Engineer / AI Practitioner | Surge Infotech, Boston, MA

Sept 2024 – Sept 2025

- Engineered **high-throughput spatial data ingestion pipelines and an ML site-ranking engine** using Databricks AutoML and MLflow, surfacing optimal urban development parcels for real estate clients with a 92% prediction accuracy.
- Built **FastAPI data extraction services to ingest 50M+ parcel and zoning records into AWS S3**, leveraging Databricks Auto Loader for incremental file discovery and schema inference and DLT for governed ingestion and automated quality checks.
- Orchestrated multi-stage Medallion pipelines (Bronze/Silver/Gold) using Databricks LakeFlow and DLT, **processing 50M+ spatial and zoning records with automated schema evolution and real-time quality validation**.
- Leveraged **OpenAI GPT-4 APIs to build an automated intelligence engine that synthesized complex pipeline metrics into executive-ready business reports**, accelerating C-suite strategic planning cycles by 75%.
- Presented smart city market adoption trends to executive leadership, aligning foundational model capabilities with commercial demands to **secure executive buy-in for a multi-quarter AI engineering product roadmap**.

### Business Data Analyst | UKG, India

Oct 2020 – Aug 2022

- Led business and technical requirements analysis for UKG TALK post-acquisition, **identifying key product gaps and shaping our strategic data backed product roadmaps**, successfully supported 10,000+ active enterprise users worldwide.
- Leveraged **SQL queries and Python funnel analysis to detect user onboarding drop-offs**, executing process optimizations techniques that **boosted premium feature monthly conversion rates by 29%** and **trial-to-paid conversions by 25%**.
- Developed **JIRA defect tracking dashboards monitoring production/staging issues**, enabling 4 cross-functional teams to unblock dependencies and critical bugs, **resulting a 43% improvement in phased sign-offs with 94% sprint delivery rate**.
- Designed **interactive Tableau and Power BI dashboards using advanced parameters and calculated fields** to report real-time campaign ROI and segment conversion metrics to non-technical stakeholders during executive quarterly reviews.
- Led **cross-functional UAT loops, post-production checks, and product marketing launch documentation** to control critical integration friction for high-value Fortune 500 clients, **boosting platform adoption rates by 2X within 12 months**.

## EDUCATION

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- **MS in Business Analytics and Information Systems**, University of South Florida, Tampa, Florida **May 2024**
- **BS in Computer Science and Engineering**, Gandhi Institute of Technology and Management, India **July 2020**

## SKILLS

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- **LLM & Agentic Systems:** Anthropic Claude API, OpenAI GPT-4, Gemini-3.7-flash, LangChain, LlamaIndex, MCP, Prompt Engineering, Data Extraction, Tool Orchestration, Multi-System Agent Workflows, Autonomous Agentic Orchestration.
- **RAG & Retrieval:** Hybrid Retrieval, Qdrant, Vector Search (HNSW/ANN), Embeddings (OpenAI Text-Embedding-3-1536d, Gemini Text Embedding-001-1536d, BGE-Large), Cross Encoder Ranking, Metadata Filtering, Chunking, Entity Resolution.
- **Evaluation & Reliability:** Confidence Calibration, Threshold/Margin Policy, Validation Harnesses, False-Positive Control, Human-In-The-Loop Gating, Latency & Cost Benchmarking.
- **Data Engineering & Cloud:** Python, PySpark, SQL, Java, FastAPI, Databricks (Auto Loader, DLT, Lakeflow, Medallion Architecture), Snowflake (Snowpipe, Streams & Tasks, Snowsight), MySQL, AWS S3/EC2, MLflow, AutoML, Git/Bitbucket.
- **Solution Engineering:** Technical Feasibility Analysis, Product Gap Analysis, Enterprise Application Integration, Proofs Of Concept (POCs), Solution Delivery, Stakeholder Management, Technical Documentation.

## ACCOMPLISHMENT

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- **Certifications:** Model Context Protocol ([Anthropic](#)), Claude with Amazon Bedrock ([Anthropic](#)), Claude with Google Cloud's Vertex AI ([Anthropic](#)), Building with Claude API ([Anthropic](#)), Introduction to agent skills (Anthropic), Introduction to subagents (Anthropic), Snowflake Data Engineering ([Snowflake](#)), McKinsey Forward Learning ([Mckinsey.org](#)).
- **Recognition:** Exceptional Performer Award (UKG -2022), Technical Excellence Awards (UKG -2021).
- **Leadership:** Graduate Assistant (USF-2024), Data Science Club (USF-2023), PMI Volunteer (Tampa-2022).